

Week 39: Continuous-time Markov Chains

Recall from discrete-time Markov chains

Definition 0.1. (*Discrete-time stochastic process*) A discrete-time stochastic process is a family of random variables $\{X_t : t \in T\}$ where T is discrete.

Definition 0.2. (*Discrete-time Markov chain*) A discrete-time Markov chain is a discrete-time stochastic process $\{X_t : t \in 0, 1, \dots\}$ that satisfies the *Markov property*

$$P(X_{t+1} = j \mid X_0 = i_0, \dots, X_t = i_t) = P(X_{t+1} = j \mid X_t = i_t).$$

Definition 0.3. (*One-step transition probabilities*) For a discrete-time Markov chain $\{X_t : t \in 0, 1, \dots\}$, we call

$$P_{i,j}^{t,t+1} = P(X_{t+1} = j \mid X_t = i)$$

the *one-step transition probabilities* from state i to state j at time t .

Theory

Definition 1.4. (*Continuous-time Markov Chain*) A continuous-time Markov chain (CTMC) is a stochastic process $\{X(t) : t \geq 0\}$ with discrete state space $\mathcal{S}$ that satisfies the Markov property and has stationary transition probabilities.

For all $s \geq 0$, $t > 0$, and $i, j \in \mathcal{S}$, we have

- Markov property:

$$P(X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j \mid X(s) = i).$$

- Stationary transition probabilities (time homogeneity):

$$P(X(s+t) = j \mid X(s) = i) = P(X(t) = j \mid X(0) = i).$$

Problem (warm-up)

Show that the Poisson process $\{X(t) : t \geq 0\}$ with rate λ satisfies the Markov property

$$P(X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j \mid X(s) = i).$$

Definition 1.5. (*Transition probability functions*) Let $\{X(t) : t \geq 0\}$ be a CTMC with state space $\mathcal{S}$. We call

$$P_{i,j}(t) = P(X(t) = j \mid X(0) = i), \quad t \geq 0, \quad i, j \in \mathcal{S}$$

the transition probability functions.

Theorem 1.6. (*Chapman-Kolmogorov*) For a CTMC,

$$P_{i,j}(s+t) = \sum_{k \in \mathcal{S}} P_{i,k}(s) P_{k,j}(t)$$

for all $t, s \geq 0$ and $i, j \in \mathcal{S}$.

Motivating birth-and-death processes

- One step at a time: $i \rightarrow i + 1$ (birth) or $i \rightarrow i - 1$ (death).
- When in state i : λ_i = rate of births and μ_i = rate of deaths.
- Formally, the jumps are defined using a little- o notation. F.ex.

$$P_{i,i+1}(h) = \lambda_i h + o(h) \text{ as } h \rightarrow 0^+$$

describes that over a very short time interval h , the $P(\text{one birth}) \approx \lambda_i h$.

- Similarly, the probability of remaining in the same state is

$$P_{i,i}(h) = 1 - (\lambda_i + \mu_i)h + o(h) \text{ as } h \rightarrow 0^+.$$

Definition 1.7. (*Birth-and-death process*) Let $\{X(t) : t \geq 0\}$ be a CTMC with state space $\mathcal{S} = \{0, 1, 2, \dots\}$ (or $\mathcal{S} = \{0, 1, \dots, N\}$). Then $\{X(t) : t \geq 0\}$ is called a birth-and-death process with nonnegative birth rates λ_i and death rates μ_i if $P_{i,j}(t)$ satisfies

1. $P_{i,i+1}(h) = \lambda_i h + o(h)$ as $h \rightarrow 0^+$ whenever $i + 1 \in \mathcal{S}$.
2. $P_{i,i-1}(h) = \mu_i h + o(h)$ as $h \rightarrow 0^+$ whenever $i - 1 \in \mathcal{S}$.
3. $P_{i,i}(h) = 1 - (\lambda_i + \mu_i)h + o(h)$ as $h \rightarrow 0^+$ for $i \in \mathcal{S}$.
4. $P_{i,j}(h) = o(h)$ as $h \rightarrow 0^+$ whenever $|i - j| > 1$.
5. $P_{i,j}(0) = \delta_{i,j}$.
6. $\mu_0 = 0$ and, for a finite state space, $\lambda_N = 0$.

Definition 1.8. (*Pure birth and pure death processes*)

- Pure birth process: $\mu_i = 0$ for all $i \geq 0$.
- Pure death process: $\lambda_i = 0$ for all $i \geq 0$.

Theorem 1.9. *In a birth-and-death process with birth rates λ_i and death rates μ_i , the following holds:*

- *Independent sojourn times.*
- *$S_i \sim \text{Exp}(\lambda_i + \mu_i)$ for $i \geq 0$.*
- *Jump probabilities in state i are*

$$P(i \rightarrow i - 1) = \frac{\mu_i}{\lambda_i + \mu_i} \quad \text{and} \quad P(i \rightarrow i + 1) = \frac{\lambda_i}{\lambda_i + \mu_i}.$$

Problem

Consider the birth-and-death process with $\mathcal{S} = \{0, 1, 2\}$, birth rates $\lambda_0 = 5$, $\lambda_1 = 4$ and $\lambda_2 = 0$, and death rates $\mu_1 = 3$ and $\mu_2 = 7$.

1. Draw the transition diagram.
2. Calculate expected sojourn times.
3. Calculate the probability that state 2 follows state 1.
4. Assume $X(0) = 0$. Calculate the probability that two consecutive births occur.

Problem (part 1)

A parking garage opens at 08:00 and closes at 16:00. We model the arrival of cars to the parking garage during opening hours as a Poisson process with rate $\lambda = 0.5$ cars/min. For now, assume that the garage has room for infinitely many cars.

1.
 - Calculate the probability that no car has arrived at 08:05.
 - Compute the expected number of cars that will arrive during the first 15 minutes of opening hours.
 - Given that two cars will arrive during the first 10 minutes, compute the probability that no cars arrive during the first 5 minutes.
2. Customers spend on average 100 kr for parking, independently of each other, with standard deviation of 10 kr.
 - Calculate the expected value of the total income at the garage during one day.
 - Calculate the variance of the total income at the garage during one day.

Theorem 1.10. $T \sim \text{Exp}(\lambda)$ is memoryless, meaning

$$P(T > s + t \mid T > s) = P(T > t)$$

for all $s, t \geq 0$.

Remark: It is the only memoryless continuous distribution on $(0, \infty)$.

Theorem 1.11. If $T_i \sim \text{Exp}(\alpha_i)$ for $i = 1, \dots, n$, and $T_1, \dots, T_n$ are independent, then

$$\min(T_1, \dots, T_n) \sim \text{Exp}\left(\sum_{i=1}^n \alpha_i\right).$$

Problem (part 2)

Assume additionally that cars, independently of each other and their arrival times, spend a stochastic time in the parking garage, which follows an exponential distribution with expectation $1/\mu = 30$ minutes. After the time has passed, the car immediately exits the parking garage.

Further, the parking garage has a maximum capacity of $N_{\max} = 20$ cars, and cars that arrive when the garage is full will drive past the garage without forming a queue. Under these assumptions, the number of cars $X(t)$ in the garage at time t is a birth-and-death process.

1.
 - Determine the birth and death rates of the process.
 - Draw a transition diagram for the process.
2. Ignore the opening hours and assume that the parking garage is always open. There are currently 16 cars in the parking garage.
 - Determine the distribution of the time until the number of cars in the garage changes.
 - Calculate the probability that the next time the number of cars in the garage changes, it will be a car that leaves.

Towards the Kolmogorov forward equations

Consider a pure birth process with $\lambda_0 \neq \lambda_1$. Derive $P_{0,1}(t)$.

Integrating factor. For ODE/IVP

$$\frac{dy}{dx} + P(x)y = Q(x)$$

we multiply each side by $\mu(x) = \exp(\int P(x) dx)$ and use the product rule.

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Theorem 1.12. (*Forward Kolmogorov differential equations*) For a birth-and-death process on $\mathcal{S} = \{0, 1, 2, \dots\}$, the $P_{i,j}(t)$ satisfy

$$P'_{i,0}(t) = -\lambda_0 P_{i,0}(t) + \mu_1 P_{i,1}(t), \quad t \geq 0$$

$$P'_{i,j}(t) = \lambda_{j-1} P_{i,j-1}(t) - (\lambda_j + \mu_j) P_{i,j}(t) + \mu_{j+1} P_{i,j+1}(t), \quad t \geq 0, \quad j \geq 1,$$

with initial conditions $P_{i,j}(0) = \delta_{i,j}$.

For $\mathcal{S} = \{0, 1, \dots, N\}$, the upper-boundary equation is $P'_{i,N}(t) = \lambda_{N-1} P_{i,N-1}(t) - \mu_N P_{i,N}(t)$.

Problem

A machine works (1) or is broken (0). Time to breakdown is $\text{Exp}(\mu)$, and time to repair is $\text{Exp}(\lambda)$. All sojourn times are independent. The machine works at time $t = 0$. Calculate the probability that the machine works at time $t = 10$.

Problem

A pure birth process starting from $X(0) = 0$ has birth parameters $\lambda_0 = 1$, $\lambda_1 = 3$, $\lambda_2 = 2$ and $\lambda_3 = 5$. Let W_1 , W_2 and W_3 be the stochastic times it takes the process to reach states 1, 2 and 3, respectively.

1. Determine the transition probability functions $P_n(t) = P(X(t) = n | X(0) = 0)$ for $n = 0, 1, 2, 3$.
2. Write W_3 as a sum of sojourn times and thereby deduce that the mean time is $\mathbb{E}[W_3] = 11/6$.
3. Determine the mean of $W_1 + W_2 + W_3$.
4. Determine the variance of W_3 .

Problem (Exam 2025)

A student at a lecture is at any time either focusing on the lecture or not. Assume that the expected length of the focused periods is 5 minutes and that the expected length of the unfocused periods is 1 minute.

- 1. Explain that a (simplified) model of the student is as a two-state continuous time Markov chain X with state space $\{0, 1\}$.
- 2. Determine the infinitesimal matrix A of X .
- 3. Assume the student is focused at the start of the lecture. What is the probability of the student being focused 10 minutes later?

Assume that a group of $n = 40$ students is modeled identically as above by $X_1, \dots, X_n$ independent two-state continuous Markov chains.

- 1. Prove that $Z = X_1 + \dots + X_n$ is a Markov chain.
- 2. Determine a differential equation for the probability distribution of Z .
- 3. Assume all students are focused at the beginning of the lecture. What is the expected number of focused students 10 minutes later?